

Markscheme

Mathematics: analysis and approaches
Practice question bank

118. (a)

8 different books are to be arranged in a row on a shelf.

Find the number of arrangements in which 2 particular books are not next to each other.

total arrangements: $8! = 40\,320$ *(M1)*

arrangements with the 2 books together (treat as one block): $7! \times 2 = 10\,080$ *(M1)* **A1**

subtract: $40\,320 - 10\,080$ *(M1)*

$= 30\,240$ **A1**

[5 marks]